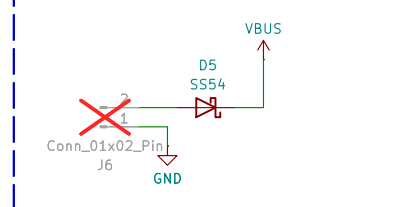
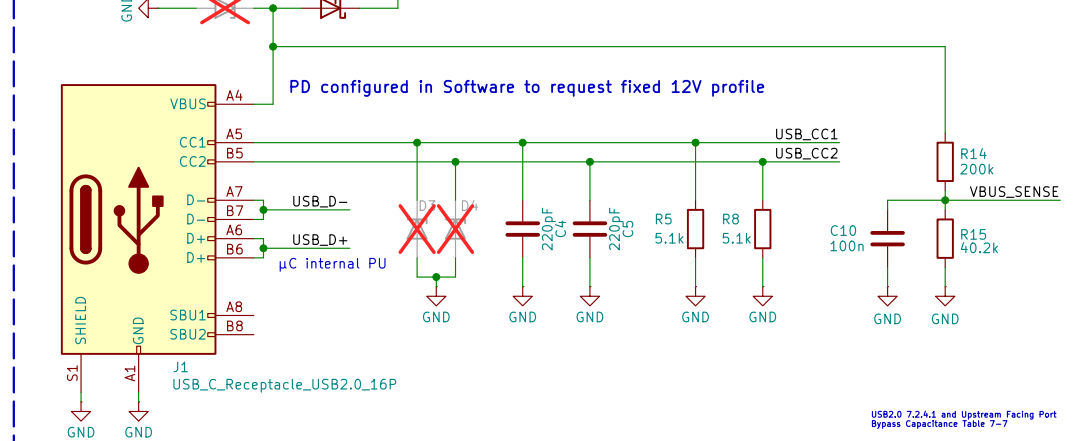


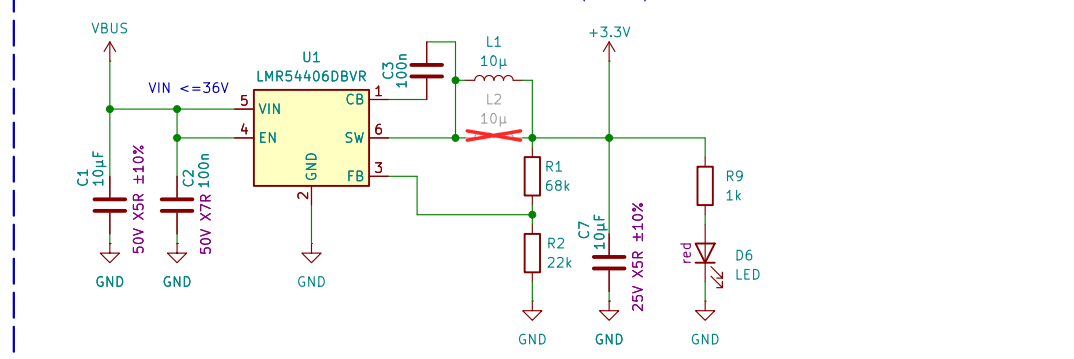
Board Power Supply (+12V)



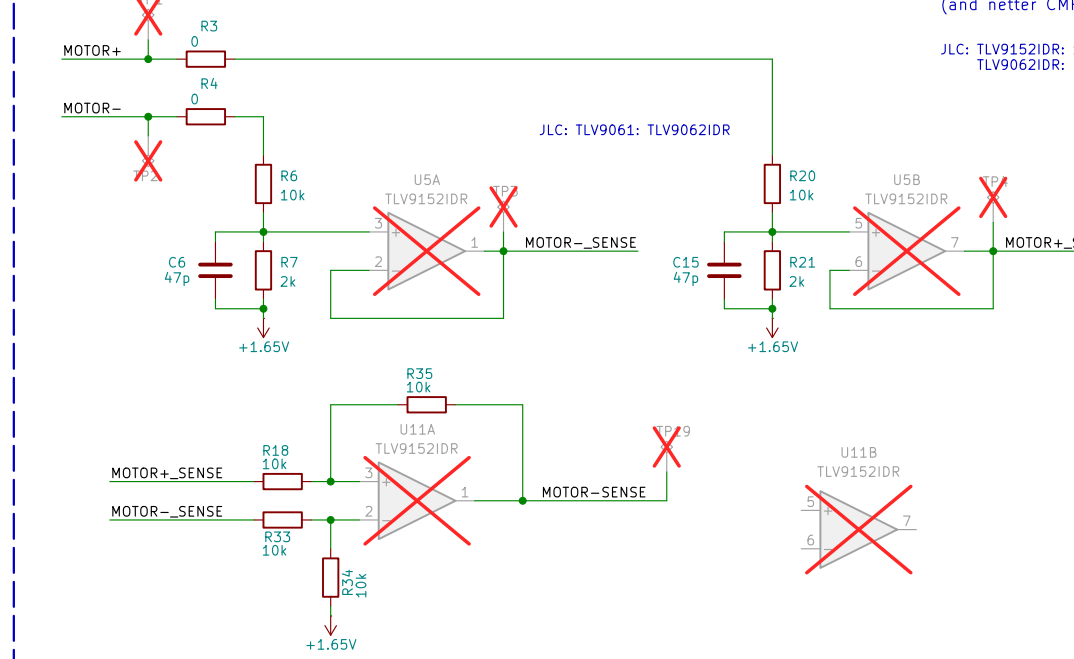
USB Data onboard $\mu C \leftrightarrow PC$. USBC Power OPTIONAL
NOT used for first experiments
only if we want to power the Board with 9/12V via USBC-PD later
(not sure if usbcdp negotiation has trouble with D2/D5)



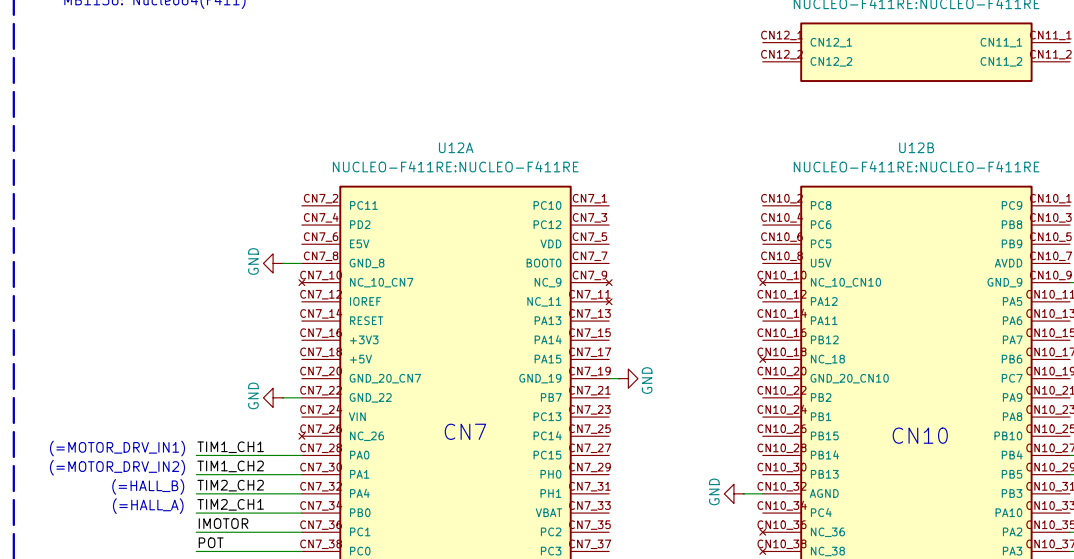
12V $\rightarrow$ 3.3V
-> potentially change to switcher LMR54406DBVR
Inductor alternatives: ASP1-40205-100M-T (C3222037), SDCL1V4020-100M-R 74405042100, XAL4040-103ME (C5144883)
handsolder: SPM4020T-100M-LR, ANR4020T100M (C6808085)



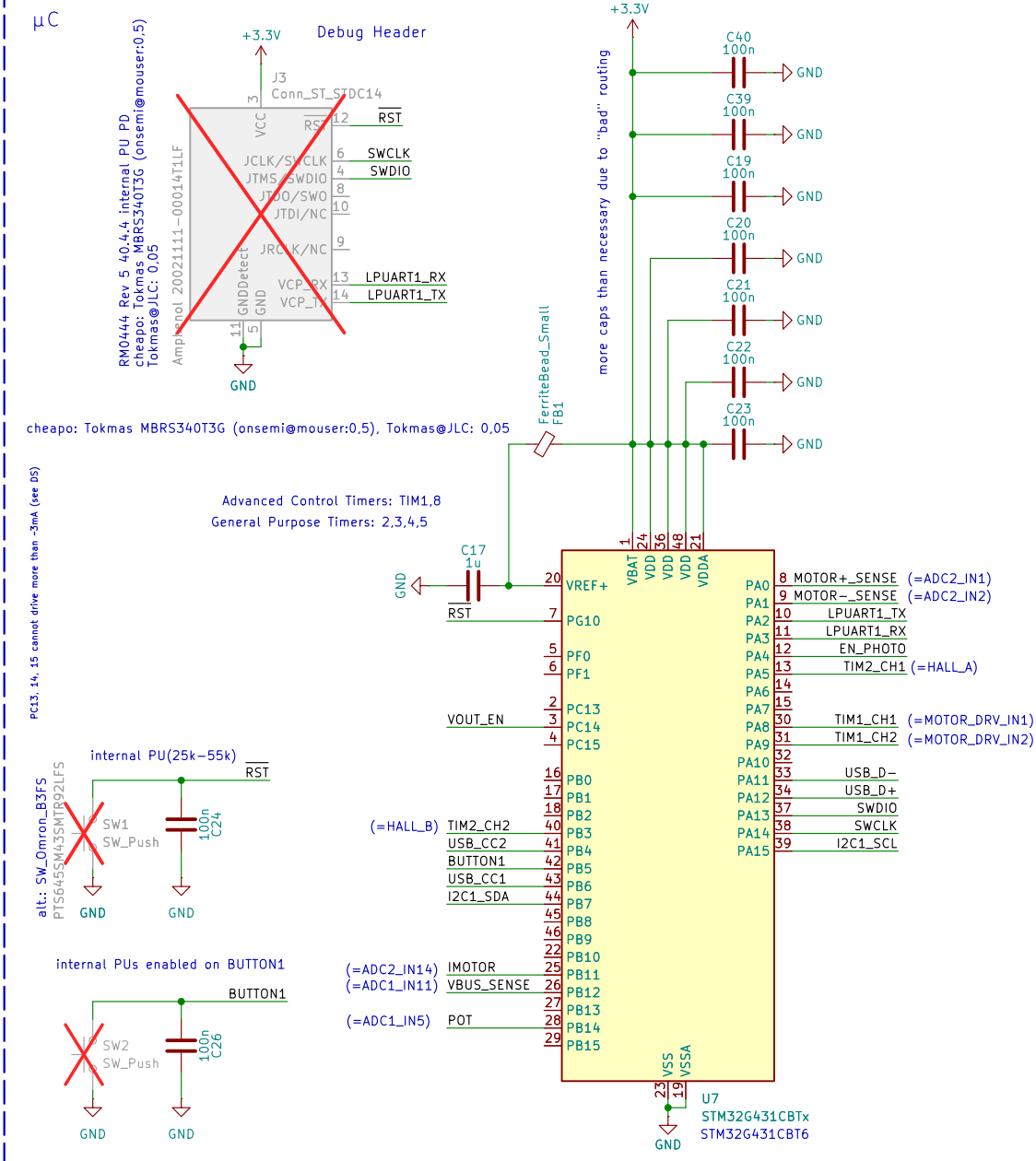
Back-EMF measurement



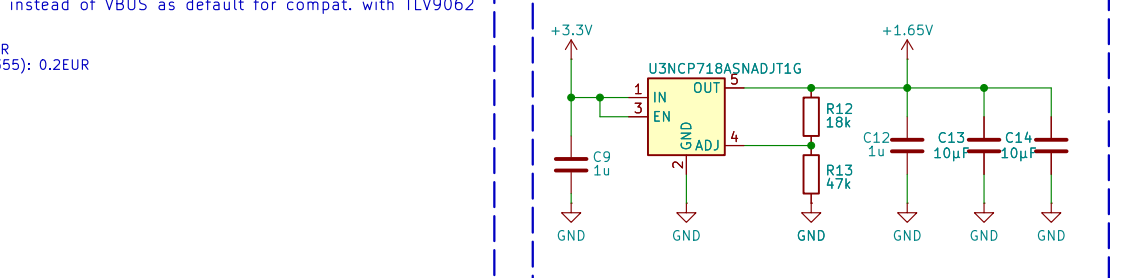
MATLAB



μC



Virtual GND for Back-EMF Measurement



5V for HALL-Supply and Back-EMF Measurement

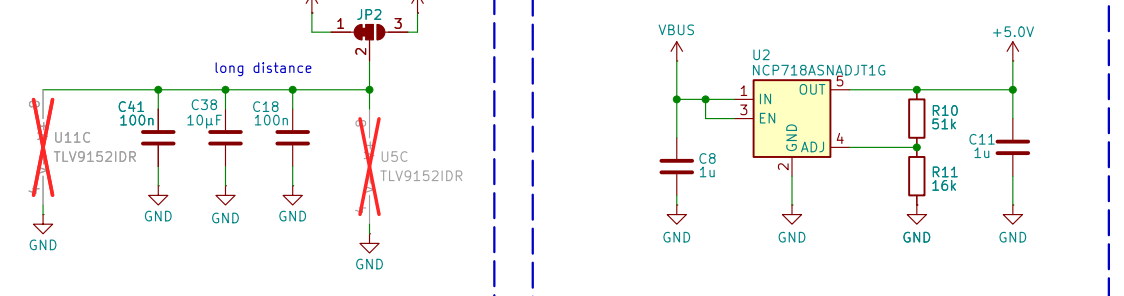


Photo Interruptor

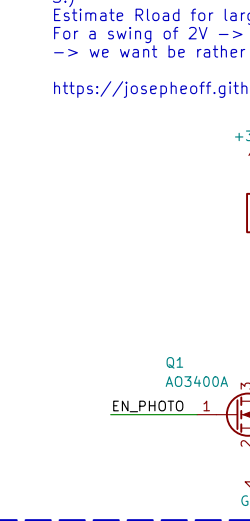
No CTR given in Datasheet

1.)
At IF = 20mA $\rightarrow$ Datasheet guarantees $> 0.5mA$ photocurrent $\rightarrow$ estimated worst case CTR $-0.5/20 = 2.5\%$
 $\rightarrow$ worst case IC @ 10mA LED drive = 240 μA
 $\rightarrow$ For Rload = 1k $\rightarrow$ Vlow.out = 3.3-240 μA *1k = 3.1V $\rightarrow$ only 0.2V swing

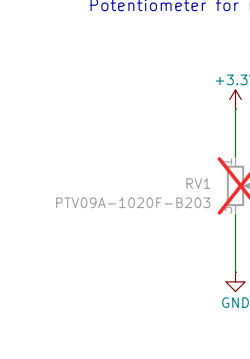
2.)
To guarantee saturation in worst case, Rload must limit Ic to 240 μA
Rload $> (3.3-V_{CE,sat})/240\mu A = 13k\Omega$

3.)
Estimate Rload for large enough output swing:
For a swing of 2V $\rightarrow$ Rload = 2/240 $\mu A = 8.3k\Omega$
 $\rightarrow$ we want be rather fast, so try with a smaller Rload first

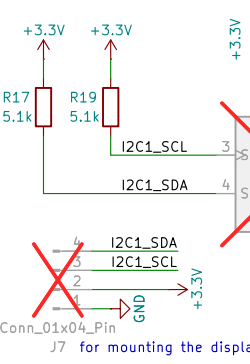
https://josephehoff.github.io/posts/rpm



Potentiometer for manual speed control



Display (optional)

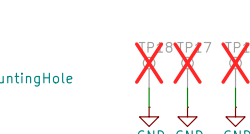


Motor Connector

pins mapped to allow TIM2 to be configured in CubeMX as "combined channels" in Encoder Mode to measure Motor Hall Encoder Signals as quadrature signals



dummy symbol to include motor mechanical step



Achtung:
Für $< 12V$: C15850 statt C440198
C1, C29, C30, C37 da für 12V ok und günstiger

Rev: V1.0
Id: 1/1

Schmitt-Trigger

Inverting config
 $\rightarrow$ for Rthevenin $< <$ Rfeedback : hysteresis-width - independent from threshold center
 $\rightarrow$ source impedance does not influence thresholds

with 1.5V swing from photointerruptor we expect 3.3V (high) $\rightarrow$ 1.8V (low)
 $\rightarrow$ set center threshold to $\sim 2.55V$
 $\rightarrow$ currently: switching threshold: 2.42V, hysteresis: 240mV

1.)
To guarantee saturation in worst case, Rload must limit Ic to 240 μA
Rload $> (3.3-V_{CE,sat})/240\mu A = 13k\Omega$

2.)
Estimate Rload for large enough output swing:
For a swing of 2V $\rightarrow$ Rload = 2/240 $\mu A = 8.3k\Omega$
 $\rightarrow$ we want be rather fast, so try with a smaller Rload first

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